

AI-Powered Retail Intelligence Digital Twin

A Deep Learning Based Decision Support System for Small and Medium Retail Businesses

TEAM MEMBERS

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KEYWORDS

Digital Twin · Deep Learning · BERT · Generative AI · Business
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Agenda

This presentation covers the full scope of the AI-Powered Retail Intelligence Digital Twin project — from research objectives and literature gaps to the technology stack and current submission status.



Objectives of the Project

Virtual Digital Twin, Forecasting Engine, Customer Intelligence, Recommendations, GenAI Copilot, What-If Simulation



Technologies & Models Used

PatchTST, XGBoost, BERT, K-Means, Apriori/FP-Growth, Llama 3/Gemma — full stack breakdown



Literature Reviewed & Gaps Identified

Six research domains surveyed; six critical gaps identified in existing SMB retail intelligence solutions



Status of Paperwork

· 52% overall completion · IEEE / Springer LNCS · Submission target: December 2026

Objectives of the Project



Virtual Retail Digital Twin

Create a store mirror to simulate conditions and preview outcomes.



Intelligent Forecasting Engine

Use PatchTST and XGBoost for demand and restock prediction.



Customer Intelligence

Segment customers, predict churn, and analyze sentiment.



Smart Recommendation System

Recommend products with collaborative filtering and basket analysis.



Generative AI Business Copilot

Answer questions and generate business reports.



What-If Simulation

Test pricing, festival, and inventory scenarios before decisions.

Intelligent Decision Support Modules

Sentiment Analysis

Flags satisfaction and complaints from reviews.

Market Basket Analysis

Finds products commonly bought together.

Recommendation Engine

Suggests products using customer behavior.

AI Business Copilot

Answers questions and supports decisions.

AI Report Generator

Creates concise business reports automatically.

How Our AI Modules Help Retail Managers

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Digital Twin

Virtual copy of the store for safe testing.

The manager simulates a 20% discount before launch.



Sales Forecasting (PatchTST)

Predicts future demand from sales patterns.

The system expects umbrella sales to rise 40% next week.

How Our AI Modules Help Retail Managers (Part 1.5)



Inventory Optimization (XGBoost)

Determines when products should be reordered and how much stock is required.

Only 18 bags of rice remain. The system predicts demand will rise and recommends ordering 100 more bags.

How Our AI Modules Help Retail Managers (Part 2)



Customer Segmentation (K-Means)

Groups customers by purchasing behaviour.

Helps managers create targeted offers.

Loyal customers get reward coupons, while new customers get welcome discounts.



Customer Churn Prediction

Identifies customers likely to stop shopping.

Helps retailers retain valuable customers early.

The system recommends sending a personalized discount coupon.

How Our AI Modules Help Retail Managers (Part 3)



Customer Footfall Prediction

Predicts how many customers will visit the store.

Helps managers schedule staff and prepare inventory.

The system predicts around 900 customers on Saturday, so the manager schedules additional billing staff.

Literature Reviewed



PatchTST: A Transformer-Based Model for Long-Term Time-Series Forecasting

Year: 2023

Focus: Sales forecasting

Contribution: Patching approach for accurate demand prediction.



Inventory Management Using Machine Learning Techniques

Year: 2022

Focus: Inventory optimization

Contribution: Used XGBoost and Random Forest for prediction.



Customer Segmentation Using RFM Analysis and K-Means Clustering

Year: 2023

Focus: Customer analytics

Contribution: Segmented customers by purchasing behaviour.

Literature Reviewed (Part 2)



Frequent Pattern Mining Using FP-Growth Algorithm

Year: 2000

Focus: Product recommendation

Contribution: Finds frequent product combinations efficiently.



Customer Churn Prediction Using XGBoost

Year: 2022

Focus: Churn prediction

Contribution: Predicts customer attrition with machine learning.



Digital Twins in Smart Manufacturing and Retail: A Survey

Year: 2024

Focus: Digital twin

Contribution: Reviews architectures for industrial and retail use.

Research Gaps Identified

PatchTST

Limitation: Forecasting only.

Gap: Add to a full retail intelligence platform.

Inventory ML

Limitation: Managed separately.

Gap: Link demand forecasting and Digital Twin.

Customer Segmentation

Limitation: Static segments.

Gap: Combine with churn and recommendations.

Research Gaps Identified (Part 2)

FP-Growth

Limitation: No personalized recommendations.

Gap: Hybrid recommendations.

Churn Prediction

Limitation: Prediction only.

Gap: AI Copilot for retention insights.

Digital Twin Survey

Limitation: Enterprise-focused, no predictive AI.

Gap: Affordable AI Digital Twin for SMBs.

Gaps Identified

Fragmented Solutions

No unified SMB platform combines forecasting, segmentation, sentiment, recommendations, and simulation.

No What-If Simulation

No SMB tool offers a Digital Twin sandbox to test decisions before execution.

Missing GenAI Copilot

Dashboards are static; there is no conversational LLM copilot for business Q&A and reports.

Cost Barrier for SMBs

Enterprise platforms are too expensive, while open-source tools lack integrated intelligence.

Reactive, Not Proactive

Most tools explain past data instead of helping teams act before problems happen.

Manual Reporting

Retail reporting is still compiled manually across disconnected data sources.

Technologies & Models Used

Module	Models	Dataset
Sales Forecasting	PatchTST, PyTorch	Rossmann
Inventory Optimization	XGBoost, Scikit-Learn	UCI Online Retail II
Customer Segmentation	K-Means, RFM	Olist E-Commerce
Sentiment Analysis	BERT, Transformers	Olist Reviews
Recommendations	Apriori, FP-Growth	Instacart
AI Copilot	Llama 3 / Gemma	—

Technology Stack

Open-source, locally deployable, and built for private on-premise LLM inference.

BACKEND & ML

Runtime

Python 3.11 · Flask

Deep Learning

PyTorch · HuggingFace

Classical ML

Scikit-Learn · XGBoost

LLM Stack

Ollama · LangChain

FRONTEND

HTML5 · CSS · JavaScript · Chart.js · D3.js

PUBLIC DATASETS

Rossmann

Demand forecasting.

Olist

Segmentation and sentiment.

Instacart

Recommendations.

UCI Retail II

Inventory optimization.

Status of Paperwork

-52%

Overall Completion

Across all milestones

15+

Papers Surveyed

Literature review complete

Dec '...

Submission Target

IEEE / Springer LNCS

Milestone	Status	Progress
Abstract & Problem Statement	✓ Done	100%
Literature Survey (15+ papers)	✓ Done	100%
System Architecture & Methodology	🔄 In Progress	65%
ML Model Implementation	🔄 In Progress	55%
Digital Twin + LLM Integration	🔄 In Progress	35%
Results & Evaluation	⌚ Pending	10%
Final Paper Draft & Submission	⌚ Pending	0%